

EX. NO: 1

CAESAR CIPHER

Aim:

To write a C program to implement the Caesar Cipher, a classical substitution technique, for encrypting a given plaintext using a fixed key value.

Algorithm:

1. Start the program.
2. Read the plaintext and the key (shift value) from the user.
3. Traverse each character of the plaintext.
4. If the character is an uppercase letter, shift it circularly within 'A' to 'Z' by the key value.
5. If the character is a lowercase letter, shift it circularly within 'a' to 'z' by the key value.
6. If the character is not an alphabet, leave it unchanged.
7. Display the resulting encrypted text.
8. Stop the program.

Program:

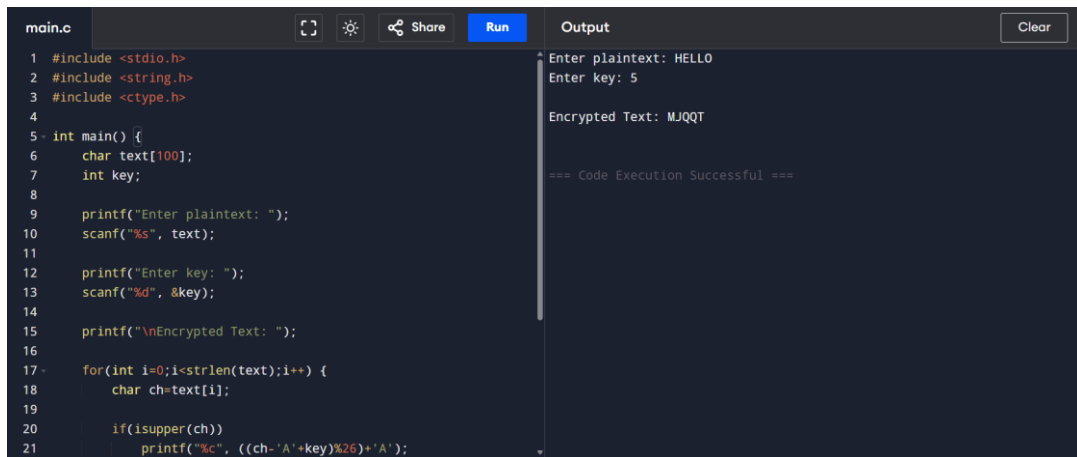
```
#include <stdio.h>
#include <string.h>
#include <ctype.h>

int main() {
    char text[100];
    int key;

    printf("Enter plaintext: ");
    scanf("%s", text);
    printf("Enter key: ");
    scanf("%d", &key);

    printf("\nEncrypted Text: ");
    for (int i = 0; i < strlen(text); i++) {
        char ch = text[i];
        if (isupper(ch))
            printf("%c", ((ch - 'A' + key) % 26) + 'A');
        else if (islower(ch))
            printf("%c", ((ch - 'a' + key) % 26) + 'a');
        else
            printf("%c", ch);
    }
    printf("\n");
    return 0;
}
```

Output:



```
main.c  [Copy] [Settings] [Share] [Run] [Clear]
1 #include <stdio.h>
2 #include <string.h>
3 #include <ctype.h>
4
5 int main() {
6     char text[100];
7     int key;
8
9     printf("Enter plaintext: ");
10    scanf("%s", text);
11
12    printf("Enter key: ");
13    scanf("%d", &key);
14
15    printf("\nEncrypted Text: ");
16
17    for(int i=0;i<strlen(text);i++) {
18        char ch=text[i];
19
20        if(isupper(ch))
21            printf("%c", ((ch-'A'+key)%26)+'A');
```

Enter plaintext: HELLO
Enter key: 5
Encrypted Text: MJQQT
=== Code Execution Successful ===

Result:

Thus a C program to implement the Caesar Cipher encryption technique was written, executed and the output was verified successfully.